

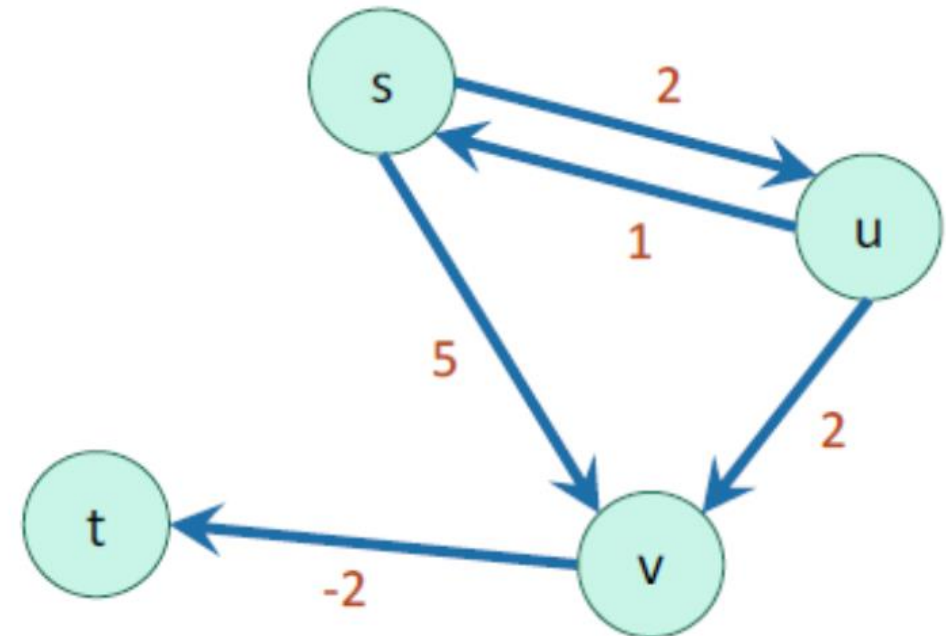
# Floyd-Warshall Algorithm (All Pairs Shortest Path)

# Floyd-Warshall

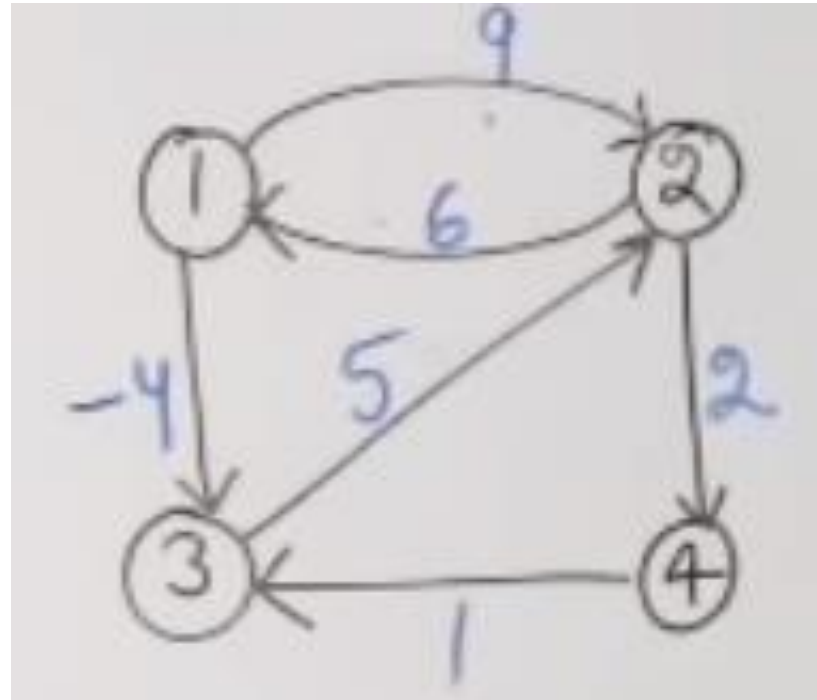
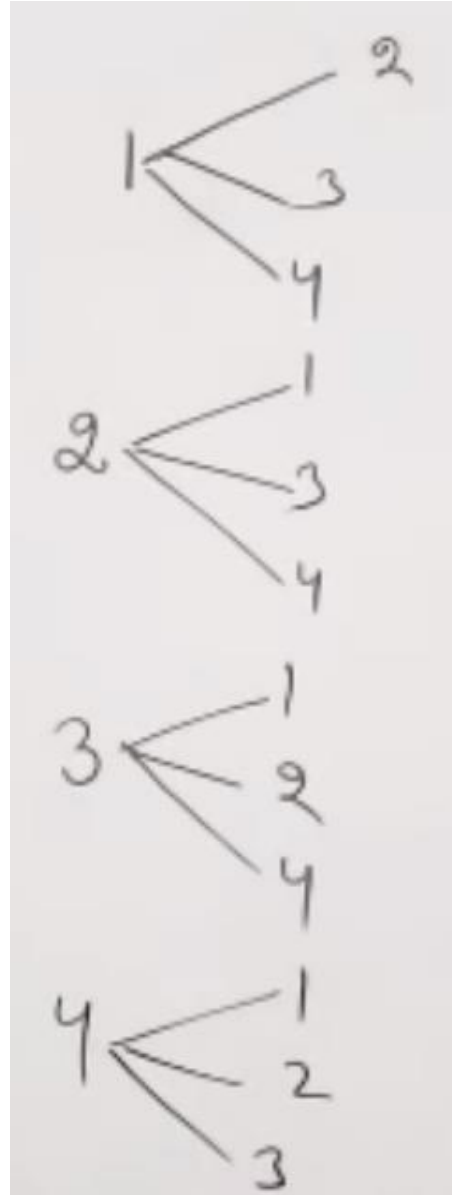
- **Floyd-Warshall Algorithm**

- This is an algorithm for **All-Pairs Shortest Paths (APSP)**
  - That is, I want to know the shortest path from  $u$  to  $v$  for **ALL pairs**  $u, v$  of vertices in the graph.
  - Not just from a special single source  $s$ .

| Source | Destination |          |          |          |    |
|--------|-------------|----------|----------|----------|----|
|        | s           | u        | v        | t        |    |
|        | s           | 0        | 2        | 4        | 2  |
|        | u           | 1        | 0        | 2        | 0  |
|        | v           | $\infty$ | $\infty$ | 0        | -2 |
|        | t           | $\infty$ | $\infty$ | $\infty$ | 0  |



# Floyd-Warshall

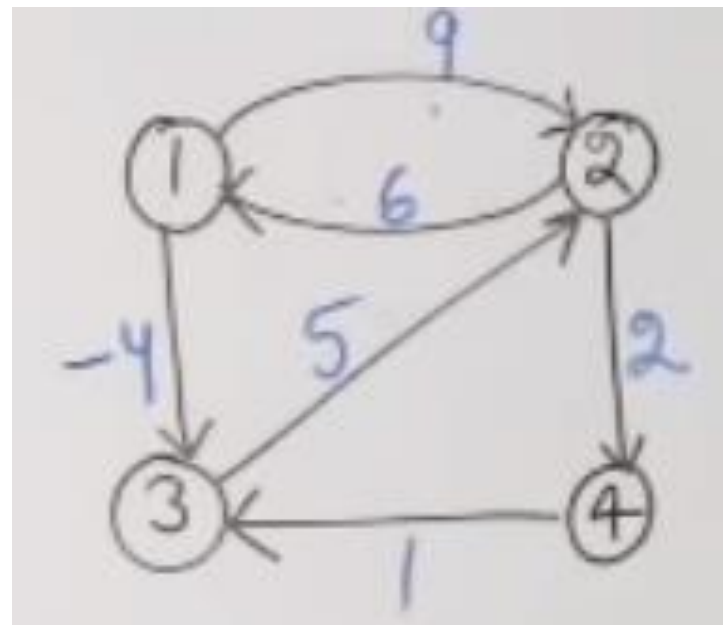


# Floyd-Warshall

Adjacency Matrix/Distance Matrix

$$D^0 = \begin{matrix} & \begin{matrix} 1 & 2 & 3 & 4 \end{matrix} \\ \begin{matrix} 1 \\ 2 \\ 3 \\ 4 \end{matrix} & \begin{bmatrix} & & & \\ & & & \\ & & & \\ & & & \end{bmatrix} \end{matrix}$$

# Floyd-Warshall



$$D^0 = \begin{matrix} & \begin{matrix} 1 & 2 & 3 & 4 \end{matrix} \\ \begin{matrix} 1 \\ 2 \\ 3 \\ 4 \end{matrix} & \left[ \begin{array}{cccc} & & & \end{array} \right] \end{matrix}$$

$$D^0 = \begin{matrix} & \begin{matrix} 1 & 2 & 3 & 4 \end{matrix} \\ \begin{matrix} 1 \\ 2 \\ 3 \\ 4 \end{matrix} & \left[ \begin{array}{cccc} 0 & 9 & -4 & \infty \\ 6 & 0 & \infty & 2 \\ \infty & 5 & 0 & \infty \\ \infty & \infty & 1 & 0 \end{array} \right] \end{matrix}$$

# Floyd-Warshall

How can we find  $D^{(k)}[u,v]$  using  $D^{(k-1)}$ ?

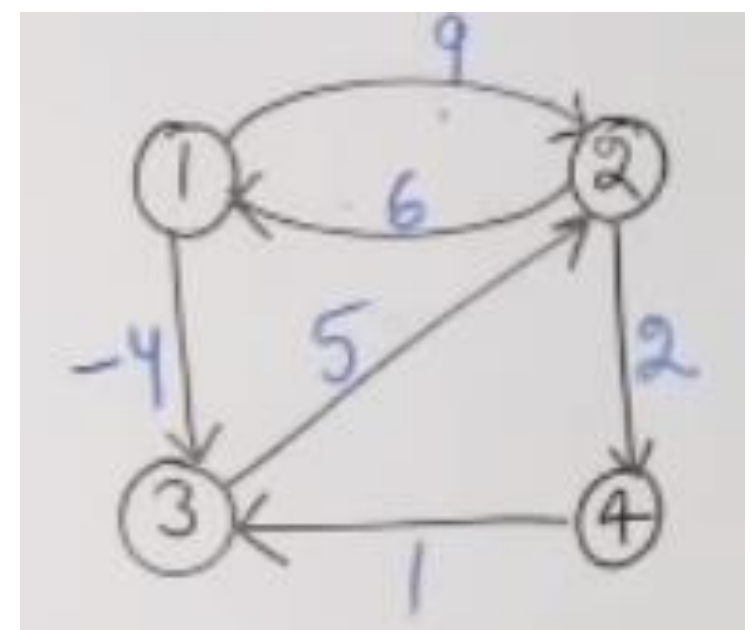
- $D^{(k)}[u,v] = \min\{ D^{(k-1)}[u,v], D^{(k-1)}[u,k] + D^{(k-1)}[k,v] \}$

**Case 1:** Cost of  
shortest path  
through  $\{1, \dots, k-1\}$

**Case 2:** Cost of shortest path  
from  $u$  to  $k$  and then from  $k$  to  $v$   
through  $\{1, \dots, k-1\}$

# Floyd-Warshall

- $D^{(k)}[u,v] = \min\{ D^{(k-1)}[u,v], D^{(k-1)}[u,k] + D^{(k-1)}[k,v] \}$



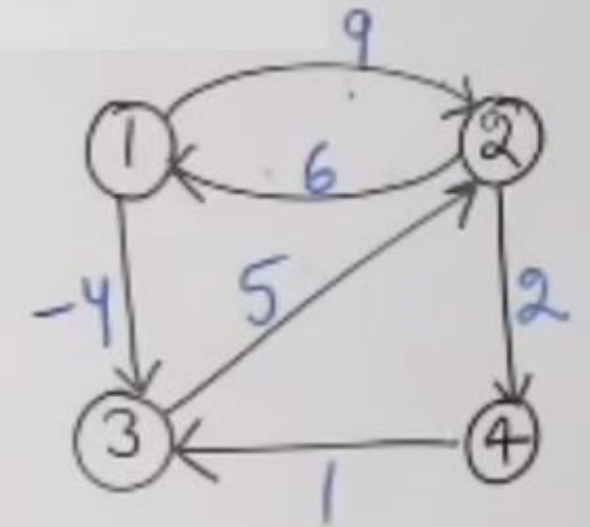
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# Floyd-Warshall

$$D^1 = \begin{matrix} & \begin{matrix} 1 & 2 & 3 & 4 \end{matrix} \\ \begin{matrix} 1 \\ 2 \\ 3 \\ 4 \end{matrix} & \begin{bmatrix} 0 & 9 & -4 & \infty \\ 6 & 0 & & \\ \infty & & 0 & \\ \infty & & & 0 \end{bmatrix} \end{matrix}$$

$$D^1 = \begin{matrix} & \begin{matrix} 1 & 2 & 3 & 4 \end{matrix} \\ \begin{matrix} 1 \\ 2 \\ 3 \\ 4 \end{matrix} & \begin{bmatrix} 0 & 9 & -4 & \infty \\ 6 & 0 & 2 & 2 \\ \infty & 5 & 0 & \infty \\ \infty & \infty & 1 & 0 \end{bmatrix} \end{matrix}$$



$$\begin{aligned} D^0[2,3] &= D^0[2,1] + D^0[1,3] \\ \infty &> 6 + (-4) = 2 \\ D^0[2,4] &= D^0[2,1] + D^0[1,4] \\ 2 &< 6 + \infty = \infty \\ D^0[3,2] &= D^0[3,1] + D^0[1,2] \\ 5 &< \infty + 9 = \infty \end{aligned}$$

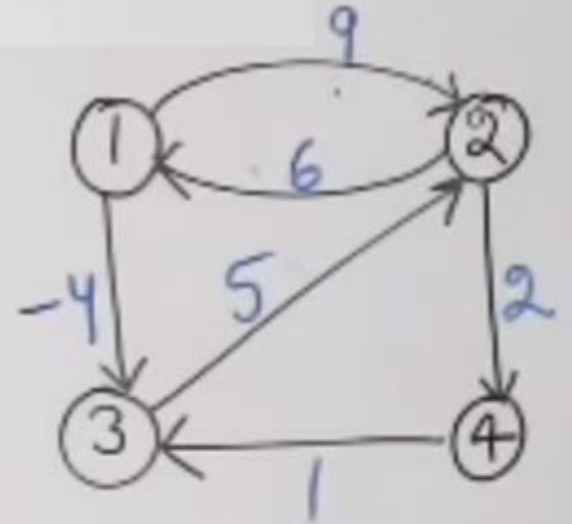
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# Floyd-Warshall

$$D^1 = \begin{matrix} & \begin{matrix} 1 & 2 & 3 & 4 \end{matrix} \\ \begin{matrix} 1 \\ 2 \\ 3 \\ 4 \end{matrix} & \begin{bmatrix} 0 & 9 & -4 & \infty \\ 6 & 0 & 2 & 2 \\ \infty & 5 & 0 & \infty \\ \infty & \infty & 1 & 0 \end{bmatrix} \end{matrix}$$

$$D^2 = \begin{matrix} & \begin{matrix} 1 & 2 & 3 & 4 \end{matrix} \\ \begin{matrix} 1 \\ 2 \\ 3 \\ 4 \end{matrix} & \begin{bmatrix} 0 & 9 & -4 & 11 \\ 6 & 0 & 2 & 2 \\ 11 & 5 & 0 & 7 \\ \infty & \infty & 1 & 0 \end{bmatrix} \end{matrix}$$



$$D^0 = \begin{matrix} & \begin{matrix} 1 & 2 & 3 & 4 \end{matrix} \\ \begin{matrix} 1 \\ 2 \\ 3 \\ 4 \end{matrix} & \begin{bmatrix} 0 & 9 & -4 & \infty \\ 6 & 0 & \infty & 2 \\ \infty & 5 & 0 & \infty \\ \infty & \infty & 1 & 0 \end{bmatrix} \end{matrix}$$

# Floyd-Warshall

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$$D^3 = \begin{matrix} & \begin{matrix} 1 & 2 & 3 & 4 \end{matrix} \\ \begin{matrix} 1 \\ 2 \\ 3 \\ 4 \end{matrix} & \begin{bmatrix} 0 & 1 & -4 & 3 \\ 6 & 0 & 2 & 2 \\ 11 & 5 & 0 & 7 \\ 12 & 6 & 1 & 0 \end{bmatrix} \end{matrix}$$

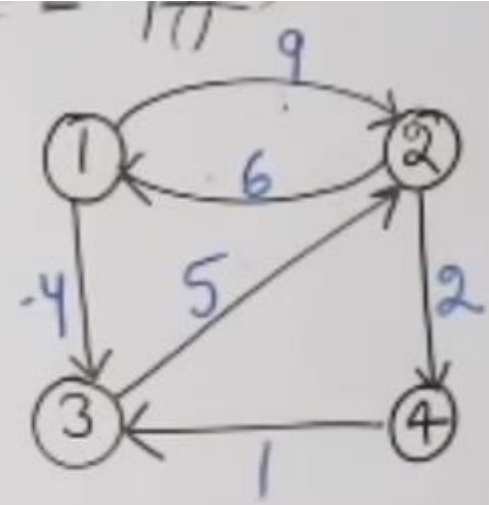
$$D^4 = \begin{matrix} & \begin{matrix} 1 & 2 & 3 & 4 \end{matrix} \\ \begin{matrix} 1 \\ 2 \\ 3 \\ 4 \end{matrix} & \begin{bmatrix} 0 & 1 & -4 & 3 \\ 6 & 0 & 2 & 2 \\ 11 & 5 & 0 & 7 \\ 12 & 6 & 1 & 0 \end{bmatrix} \end{matrix}$$

$$D^3[1,2] = D^3[1,4] + D^3[4,2]$$

$$-4 < 3 + 6$$

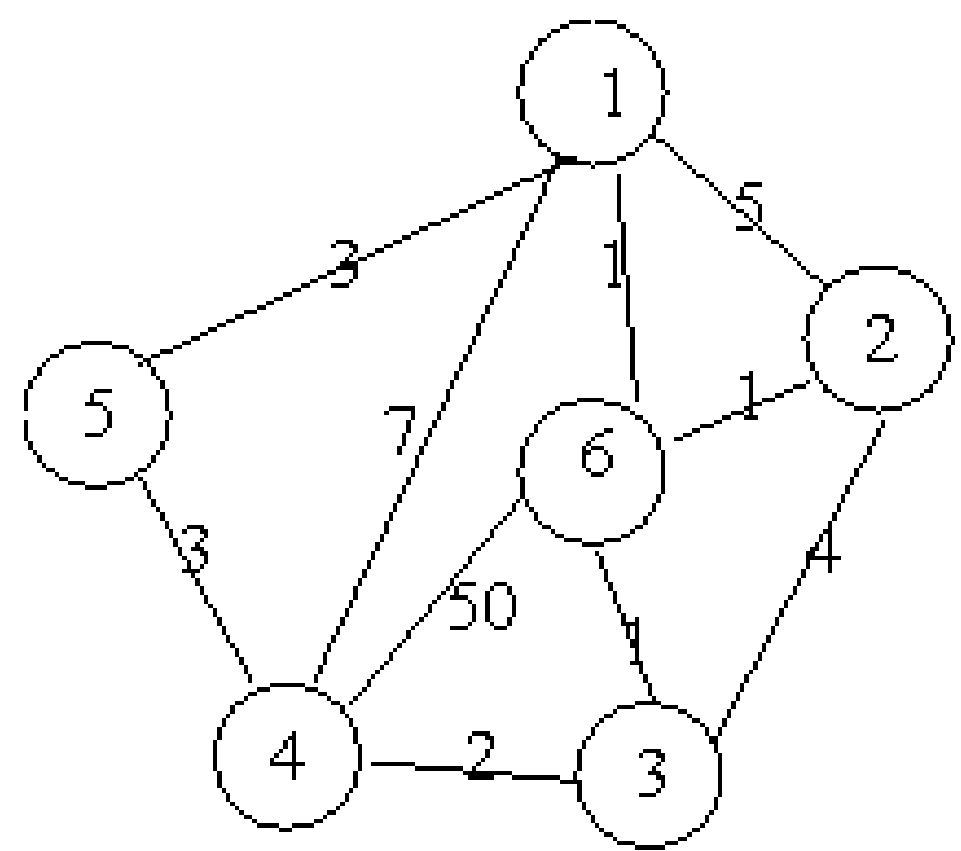
$$D^3[1,3] = D^3[1,4] + D^3[4,3]$$

$$-4 < 3 + 1$$



$$D^0 = \begin{matrix} & \begin{matrix} 1 & 2 & 3 & 4 \end{matrix} \\ \begin{matrix} 1 \\ 2 \\ 3 \\ 4 \end{matrix} & \begin{bmatrix} 0 & 9 & -4 & \infty \\ 6 & 0 & \infty & 2 \\ \infty & 5 & 0 & \infty \\ \infty & \infty & 1 & 0 \end{bmatrix} \end{matrix}$$

# Floyd-Warshall



# Floyd-Warshall

FLOYD-WARSHALL( $W$ )

1.  $n \leftarrow \text{rows}[W]$
2.  $D^{(0)} \leftarrow W$
3. **for**  $k \leftarrow 1$  **to**  $n$
4.     **do for**  $i \leftarrow 1$  **to**  $n$
5.         **do for**  $j \leftarrow 1$  **to**  $n$
6.              $d_{ij}^{(k)} \leftarrow \min(d_{ij}^{(k-1)}, d_{ik}^{(k-1)} + d_{kj}^{(k-1)})$
7. **return**  $D^{(n)}$

- Running time:  $O(n^3)$ 
  - Better than running BF  $n$  times!
  - Not really better than running Dijkstra  $n$  times.
    - But it's simpler to implement and handles negative weights.